Fm 16
Leon McQueen

Diff 7 HW

2/25/26

<u>Review</u>

1) $p = Aq^n$

$\log p = n \log q + \log A$

$\therefore n = -2$

$\log A = 0.5 \rightarrow A = \sqrt{10} = 3.2$

2 a

3 $y = \ln(2-3x)$

$y' = -\frac{3}{2-3x}$

when $x=0$, $y' = -\frac{3}{2}$

4 stretch vertically parallel to y axis by scale factor 2.

translate by $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

5 $y = \frac{x^{\frac{1}{2}}}{1+x^{\frac{1}{2}}}$

$y' = \frac{\frac{1}{2}x^{-\frac{1}{2}} \times (1+x^{\frac{1}{2}}) - \frac{1}{2}x^{-\frac{1}{2}} \times x^{\frac{1}{2}}}{(1+x^{\frac{1}{2}})^2}$

$= \frac{\left(\frac{1}{2\sqrt{x}} + \frac{1}{2}\right) - \frac{1}{2}}{1 + 2\sqrt{x} + x}$

$= \frac{1}{2\sqrt{x}(1+\sqrt{x})^2}$ c

<u>Consolidation</u>

1 i) $3y^2 \frac{dy}{dx}$

ii) $\frac{dy}{dx} \cos y$

iii) $\frac{dy}{dx} e^y$

iv) $\frac{1}{y} \frac{dy}{dx}$

$2 \quad y^4 + 2xy = x^2$

$4y^3\frac{dy}{dx} + \left(2y + 2x\frac{dy}{dx}\right) = 2x$

$\frac{dy}{dx}\left(4y^3 + 2x\right) + 2y = 2x$

$\frac{dy}{dx} = \frac{2x - 2y}{4y^3 + 2x}$

$= \frac{x - y}{2y^3 + x}$

$3 \quad y^2 + 2xy\frac{dy}{dx}$

$\sin y + \frac{dy}{dx}x\cos y$

$\frac{(1+x)\frac{dy}{dx} - y}{(1+x)^2}$

$4 \quad 3x^2 + 3y^2\frac{dy}{dx} - 3 = 0$

$3x^2 - 3 = 0$

$x = 1 \quad or \quad x = -1$

$\downarrow \qquad \downarrow$

$y = 2 \quad or \quad y = \sqrt[3]{2}$

$5 \quad y^2 = \frac{x^2}{1+2x}$

$2y\frac{dy}{dx} = \frac{2x(1+2x) - 2x^2}{(1+2x)^2}$

$\frac{dy}{dx} = \frac{2x + 2x^2}{2y(1+2x)^2}$

$= \frac{x(1+x)}{y(1+2x)^2}$

$6 \quad y = 2^x$

$\ln y = x\ln 2 \qquad k = \ln 2$

$\frac{1}{y}\frac{dy}{dx} = \ln 2$

$\frac{dy}{dx} = y\ln 2$

$= 2^x \ln 2$

7 $x^2 + kxy + 4y^2 = 48$

$2x + ky + \frac{dy}{dx}kx + 8y\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{2x+ky}{kx+8y}$

when $k=4$, $\frac{dy}{dx} = -\frac{2x+4y}{4x+8y}$

$= -\frac{1}{2}$

8 $\frac{dV}{dt} = 32\pi$

$\frac{dV}{dr} = 4\pi r^2$

$\frac{dr}{dt} = ?$

$32\pi = 4\pi r^2 \times \frac{dr}{dt}$

$\frac{dr}{dt} = \frac{8}{r^2}$

$= 2$ when $r = 2$

9 $h = 20$

$r = 4$

$\frac{dh}{dt}$

$\frac{dV}{dt} = 2$

$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$

$V = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi\left(\frac{h}{5}\right)^2 h$

$= \frac{1}{75}\pi h^3$

$\frac{dV}{dh} = -\frac{1}{25}\pi h^2$

$2 = -\frac{1}{25}\pi h^2 \times \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{-50}{\pi h^2} = -0.0632 \text{ cms}^{-1}$

$r = \frac{h}{5}$

$\frac{1}{75}\pi 20^3 = \frac{320}{3}\pi$

$\frac{1}{75}\pi h^3 = \frac{160}{3}\pi$

$h^3 = 4000$

$h = \sqrt[3]{4000}$

$\therefore \frac{dh}{dt} = \frac{2}{25} \times \pi \times 4000^{\frac{2}{3}}$

Exam Qs

1) $x^2-4xy+8y^3-4=0$

$2x - 4x\frac{dy}{dx} - 4y + 24y^2\frac{dy}{dx} = 0$

At stationary point:

$2x-4y=0$

$x=2y$

Then $x^2-4xy+8y^3-4=0$

$x^2-2x^2+x^3-4=0$
$x^3-x^2-4=0$
} sub in value of y @ stationary point

Which only has one real solution: $x=2$

At $x=2$, $y=\frac{x}{2}=1$.

So only 1 stationary point at $(2,1)$

2 $x\sin y+\cos 2y=\frac{5}{2}$ | $x\frac{dy}{dx}\cos y+\sin y - \frac{dy}{dx}2\sin 2y=0$

$\frac{dy}{dx}=-\frac{\sin y}{x\cos y-2\sin 2y}$

$\therefore x\cos y-2\sin 2y=0$

$x=\frac{2\sin 2y}{\cos y}=\frac{4\sin y\cos y}{\cos y}$

$\frac{4\sin^2 y\cos y}{\cos y}+\cos 2y=\frac{5}{2}$

$+2\cos^2 y-1=0$

$4\sin^2 y\cos y+2\cos^3 y-\cos y=\frac{5}{2}\cos y$

$8\cos y-8\cos^3 y+4\cos^3 y-2\cos y=5\cos y$

$-4\cos^3 y-\cos y=0$

$4\cos^3 y-\cos y=0$

$\cos y(4\cos^2 y-1)=0$

$\cos y=0$ or $\cos y=\pm\frac{\sqrt{3}}{2}$

$y: \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{3}$

$x: 3.46, 3.5, 3.46, -3.46, -3.5, -3.46$